AI-use disclosure (outside report page limit)

Code, outputs and run instructions: https://github.com/Harshu3605/quantum-traffic-gridlock-qaoa

Tool. Claude (Anthropic, model Claude Opus 5.5), used through Claude Code.

What the AI did.

- Read the problem statement.
- Proposed the formulation: exact per-approach decomposition plus an order-2 Walsh projection.
- Wrote all code in `qtraffic/`, `run.py`, `experiments.py` and `tests/`.
- Ran the experiments and drafted README.md, REPORT.md, this disclosure and the plain-language explainer (`docs/`), including the scripts that render them to PDF.

What the team did.

- Provided the problem statement and the organisers' official scorer and scenario files (R and H).
- Directed the work, chose the repository, and created and published the GitHub repo.
- Reviewed the outputs.

How results were verified independently of the AI's claims.

- All reported costs come from the organisers' `official/scorer.py`, used unmodified. Our simulator and QUBO decomposition are asserted equal to it on 40 random scenarios × 256 schedules. The four verification costs come from the PS.
- The decomposition C = Σ D_k + switches is asserted exactly on all 256 schedules (unit test).
- The QUBO↔Ising conversion is asserted equal on sample states (unit test).
- The Qiskit circuit distribution is compared to the trained numpy distribution (TVD ≈ 1e−15 on every run). A test re-imports `circuit.qasm` and compares it to `counts.json`.
- Determinism: the same seed gives identical results (unit test).
- All numbers in REPORT.md are copied from `outputs/experiments/experiments.json` and `outputs/*/seed_*/results.json`, which anyone can regenerate with the commands in README.md.

Team responsibility. The team takes responsibility for the submission and has reviewed the submitted outputs. The H outputs were produced by the unchanged pipeline right after the organisers released `scenario_H.json`.